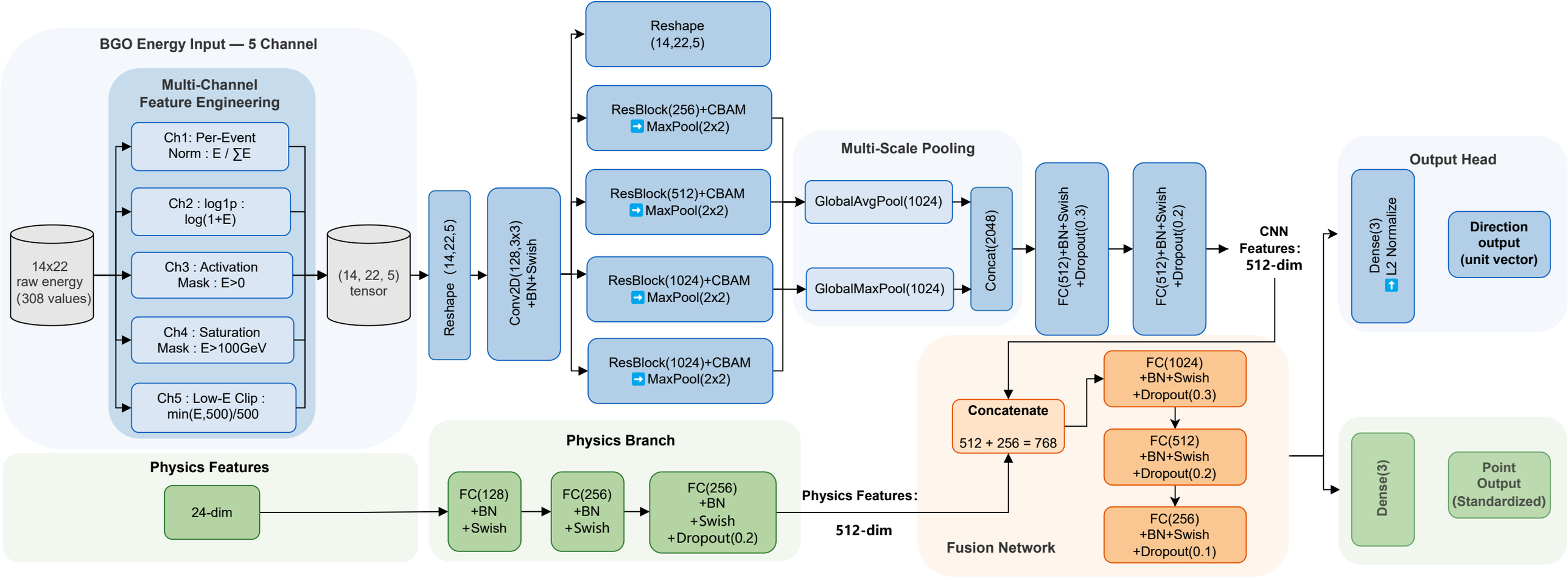

BGO Energy Input — 5 Channel
14x22
raw energy
(308 values)
Multi-Channel
Feature Engineering
Ch1: Per-Event
Norm : E / ∑E
Ch2 : log1p :
log(1+E)
Ch3 : Activation
Mask : E>0
Ch4 : Saturation
Mask : E>100GeV
Ch5 : Low-E Clip :
min(E,500)/500
(14, 22, 5)
tensor
Reshape (14,22,5)
Conv2D(128,3x3)
+BN+Swish
Reshape
(14,22,5)
ResBlock(256)+CBAM
MaxPool(2x2)
ResBlock(512)+CBAM
MaxPool(2x2)
ResBlock(1024)+CBAM
MaxPool(2x2)
ResBlock(1024)+CBAM
MaxPool(2x2)
Multi-Scale Pooling
GlobalAvgPool(1024)
GlobalMaxPool(1024)
Concat(2048)
FC(512)+BN+Swish
+Dropout(0.3)
FC(512)+BN+Swish
+Dropout(0.2)
CNN
Features:
512-dim
Output Head
Dense(3)
L2 Normalize
Direction
output
(unit vector)
Physics Features
24-dim
Physics Branch
FC(128)
+BN
+Swish
FC(256)
+BN
+Swish
FC(256)
+BN
+Swish
+Dropout(0.2)
Physics Features:
512-dim
Concatenate
512 + 256 = 768
FC(1024)
+BN+Swish
+Dropout(0.3)
FC(512)
+BN+Swish
+Dropout(0.2)
FC(256)
+BN+Swish
+Dropout(0.1)
Fusion Network
Dense(3)
Point
Output
(Standardized)